Ibn Tofail University

Analysis IV

Problem Set I

Exercise 1:

Study the convergence of the following series:

$$\sum_{n=1}^{\infty} \frac{1}{n}; \qquad \sum_{n=2}^{\infty} \frac{n^2+1}{n^2}; \qquad \sum_{n=1}^{\infty} \frac{2}{\sqrt{n}}$$

$$\sum_{n=1}^{\infty} \frac{(2n+1)^4}{(7n^2+1)^3}; \quad \sum_{n=1}^{\infty} \left(ne^{\frac{1}{n}} - n\right); \quad \sum_{n=0}^{\infty} \ln\left(1+e^{-n}\right)$$

$$\sum_{n=1}^{\infty} \left(\frac{n}{n+1}\right)^{n^2}$$

Correction

1. **Serie I:** We study the convergence of the series $\sum_{n=1}^{\infty} \frac{1}{n}$.
   (a) The general term is positive and tends to 0 as $n \to \infty$.
   (b) we use the integral test:

$$\int_1^{\infty} \frac{1}{x}\, dx = \lim_{m\to\infty} \ln(m) - \ln(1) = \infty$$

$$\boxed{\text{The series diverges.}}$$

2. **Serie II:** We study the convergence of the series $\sum_{n=2}^{\infty} \frac{n^2+1}{n^2}$.
   (a) The general term is positive and tends to 1 as $n \to \infty$.

$$\boxed{\text{The series diverges.}}$$

3. **Serie III:** We study the convergence of the series $\sum_{n=1}^{\infty} \frac{2}{\sqrt{n}}$.
   (a) The general term is positive and tends to 0 as $n \to \infty$.
   (b) we use the p-series test:

$$a_n = \frac{1}{n^{1/2}}$$

   . Here, $p = \frac{1}{2} < 1$.

$\boxed{\text{The series diverges.}}$

4. **Serie IV:** We study the convergence of the series $\sum_{n=0}^{\infty} \frac{(2n+1)^4}{(7n^2+1)^3}$.
   (a) The general term is positive and tends to 0 as $n \to \infty$.
   (b) For large n, we approximate:
   $$a_n \approx \frac{16n^4}{343n^6} = \frac{16}{343} \cdot \frac{1}{n^2}$$
   (c) Compare with the p-series $\sum \frac{1}{n^2}$, which converges since $p = 2 > 1$. By the limit comparison test:
   $$\lim_{n\to\infty} \frac{a_n}{1/n^2} = \frac{16}{343} \neq 0, \infty$$
   $$\boxed{\text{The series converges.}}$$
5. **Serie V:** We study the convergence of the series $\sum_{n=1}^{\infty} \left(ne^{\frac{1}{n}} - n\right)$.
   (a) the general term is positive and tends to 1 as $n \to \infty$.
   $$\boxed{\text{The series diverges.}}$$
6. **Serie VI:** We study the convergence of the series $\sum_{n=0}^{\infty} \ln(1 + e^{-n})$.
   (a) the general term tends to 2 as $n \to \infty$.
   $$\boxed{\text{The series diverges.}}$$
7. **Serie VII:** We study the convergence of the series $\sum_{n=1}^{\infty} \left(\frac{n}{n+1}\right)^{n^2}$.
   (a) the general term tends to 1 as $n \to \infty$
   $$\boxed{\text{The series diverges.}}$$

Exercise 2:

Determine the nature of the numerical series with the following general terms:

$$u_n = \frac{1}{n\cos^2 n}; \quad v_n = \frac{1}{[\ln(n)]^n}; \quad w_n = \frac{\tan^n\left(\frac{\pi}{7}\right)}{3^{n+2}}$$

$$x_n = \frac{1}{n\ln(n)}; \quad y_n = \frac{1}{n(\ln n)^2}; \quad z_n = \frac{1}{(\ln 2)^2 + (\ln 3)^2 + \ldots + (\ln n)^2}$$

Correction

1. We study the convergence of the series u_n.
   (a) since the general term does not converge to any number it doesnt converge to 0 too.

   $$\boxed{\text{The series diverges.}}$$

2. We study the convergence of the series v_n.
   (a) the general term is positive and tends to 0 as $n \to \infty$.
   (b) we apply **the root test**:

   $$\lim_{n\to\infty} \sqrt[n]{|a_n|} = \lim_{n\to\infty} \frac{1}{\ln(n)} = 0 < 1$$

   $$\boxed{\text{The series converges.}}$$

3. We study the convergence of the series w_n.
   (a) we know it a geometric series with ratio $r = \frac{\tan(\pi/7)}{3}$.
   (b) we apply the **geometric series test**: we have $|r| < 1$.

   $$\boxed{\text{The series converges.}}$$

4. We study the convergence of the series x_n.
   (a) the general term is positive and tends to 0 as $n \to \infty$.
   (b) we apply the **integral test**:

   $$\int_2^\infty \frac{1}{x\ln(x)}\, dx = \lim_{m\to\infty} [\ln(\ln(m)) - \ln(\ln(2))] = \infty$$

   $$\boxed{\text{The series diverges.}}$$

5. We study the convergence of the series y_n.
   (a) the general test is positive and tends to 0 as $n \to \infty$.
   (b) we apply the **integral test**:

   $$\int_2^\infty \frac{1}{x(\ln x)^2}\, dx = \frac{1}{\ln(2)} < \infty$$

   $$\boxed{\text{The series converges.}}$$

6. We study the convergence of the series z_n.
   (a) The general term is positive and tends to 0 as $n \to \infty$.
   (b) We compare with the series $\frac{1}{n(\ln n)^2}$ and apply the **comparison test**:

   $$z_n = \frac{1}{(\ln 2)^2 + (\ln 3)^2 + \cdots + (\ln n)^2} \sim \frac{1}{n(\ln n)^2}.$$

   Since $\sum \frac{1}{n(\ln n)^2}$ converges, the series z_n also converges.

   $$\boxed{\text{The series converges.}}$$

Exercise 3:

Show that the series with general term

$$w_n = \frac{(-1)^n}{\ln(1+\sqrt{n})}$$

is convergent but not absolutely convergent (called semi-convergent).

Correction

1. **Absolute convergence.** We study the convergence of the series $|w_n|$.

$$\sum_{n=1}^{\infty} |w_n| = \sum_{n=1}^{\infty} \frac{1}{\ln(1+\sqrt{n})}.$$

For large n, $\ln(1+\sqrt{n}) \sim \frac{1}{2}\ln n$, so

$$\frac{1}{\ln(1+\sqrt{n})} \sim \frac{2}{\ln n}.$$

Since $\sum 1/\ln n$ diverges, the series is **not absolutely convergent**.

2. **Conditional convergence.** We study the convergence of the series

$$a_n = \frac{1}{\ln(1+\sqrt{n})} > 0.$$

- a_n is decreasing for large n. - $\lim_{n\to\infty} a_n = 0$. By the **alternating series test (Leibniz criterion)**, the series

$$\sum_{n=1}^{\infty} (-1)^n a_n$$

converges conditionally.

$$\boxed{\text{The series is convergent (semi-convergent).}}$$

Exercise 4:

1. Show that the following two series are absolutely convergent:

$$\sum_{n=0}^{+\infty} \frac{1}{n!} \quad \text{and} \quad \sum_{n=0}^{+\infty} \frac{(-1)^n}{2^n}$$

2. Deduce that the series:

$$\sum_{n=0}^{+\infty} \left[\sum_{k=0}^{n} \frac{(-1)^{n-k}}{k! \cdot 2^{n-k}} \right]$$

is absolutely convergent and determine its value.

Correction

1. **Absolute convergence of the first two series.**

   (a) Consider $\sum_{n=0}^{\infty} \frac{1}{n!}$. Using the ratio test:

   $$\frac{a_{n+1}}{a_n} = \frac{1/(n+1)!}{1/n!} = \frac{1}{n+1} \to 0 < 1$$

   so the series converges absolutely.

   (b) Consider $\sum_{n=0}^{\infty} \frac{(-1)^n}{2^n}$. Its absolute series is geometric:

   $$\sum_{n=0}^{\infty} \frac{1}{2^n}, \quad r = \frac{1}{2} < 1$$

   so it converges absolutely.

2. **Absolute convergence of the Cauchy product.**

   (a) Let

   $$a_n = \frac{1}{n!}, \quad b_n = \frac{(-1)^n}{2^n}.$$

   The series

   $$\sum_{n=0}^{\infty} \sum_{k=0}^{n} a_k b_{n-k}$$

   is the Cauchy product of two absolutely convergent series, hence convergent.

   (b) **Determine its value.**

   i. Compute the sums:

   $$\sum_{n=0}^{\infty} a_n = \sum_{n=0}^{\infty} \frac{1}{n!} = e, \quad \sum_{n=0}^{\infty} b_n = \sum_{n=0}^{\infty} \frac{(-1)^n}{2^n} = \frac{1}{1-(-1/2)} = \frac{2}{3}.$$

   ii. By the Cauchy product formula, the sum of the series is

   $$e \cdot \frac{2}{3} = \frac{2e}{3}.$$

$$\boxed{\text{The series is absolutely convergent and its sum is } \frac{2e}{3}.}$$

Exercise 5:

Study the nature of the following series:

$$\sum_{n>2} \ln\left(1 + \frac{(-1)^n}{n}\right), \quad \sum_{n>1} \sin\left(\frac{(-1)^n}{n}\right)$$

Correction

1. Consider the series
$$\sum_{n>2} \ln\left(1+\frac{(-1)^n}{n}\right).$$
   (a) For large n, $\ln(1+x) \sim x$:
   $$\ln\left(1+\frac{(-1)^n}{n}\right) \sim \frac{(-1)^n}{n}.$$
   (b) The series $\sum \frac{(-1)^n}{n}$ is alternating with terms tending to 0, so by the **alternating series test (Leibniz criterion)**, the series converges conditionally.
   (c) Absolute convergence fails since
   $$\sum_{n>2} \left|\ln\left(1+\frac{(-1)^n}{n}\right)\right| \sim \sum_{n>2} \frac{1}{n} = \infty.$$
   $$\boxed{\text{The series converges conditionally but not absolutely.}}$$
2. Consider the series
$$\sum_{n>1} \sin\left(\frac{(-1)^n}{n}\right).$$
   (a) For small x, $\sin(x) \sim x$:
   $$\sin\left(\frac{(-1)^n}{n}\right) \sim \frac{(-1)^n}{n}.$$
   (b) Again, by the **alternating series test**, the series converges conditionally.
   (c) Absolute convergence fails since
   $$\sum_{n>1} \left|\sin\left(\frac{(-1)^n}{n}\right)\right| \sim \sum_{n>1} \frac{1}{n} = \infty.$$
   $$\boxed{\text{The series converges conditionally but not absolutely.}}$$

Exercise 6:

Show that the series with general terms

$$u_n := \frac{(-1)^n}{\sqrt{n}} \quad \text{and} \quad v_n := \frac{(-1)^n}{\sqrt{n+(-1)^n}}$$

are not of the same nature, although $u_n \sim v_n$.

Correction

1. Consider $u_n = \frac{(-1)^n}{\sqrt{n}}$.
   (a) The sequence $a_n = 1/\sqrt{n}$ is positive, decreasing, and tends to 0.
   (b) By the **alternating series test (Leibniz criterion)**, the series $\sum u_n$ converges conditionally.
   (c) Absolute convergence fails since $\sum |u_n| = \sum 1/\sqrt{n}$ diverges.
2. Consider $v_n = \frac{(-1)^n}{\sqrt{n+(-1)^n}}$.
   (a) Split into even/odd terms: $b_{2k} = 1/\sqrt{2k+1}$, $b_{2k+1} = 1/\sqrt{2k}$.
   (b) The sequence (b_n) is not monotone, so the Leibniz criterion cannot be applied.
   (c) Although $b_n \sim a_n \sim 1/\sqrt{n}$ as $n \to \infty$, the series $\sum v_n$ diverges.

The series $\sum u_n$ converges conditionally, while $\sum v_n$ diverges. Hence, they are not of the same nature, despite $u_n \sim v_n$.